Lecture 6 — Heteroskedasticity

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Reading: Chapter 8

Why revisit the error variance?

So far, we have used the homoskedasticity assumption

$$\mathrm{Var}(u_i \mid X) = \sigma^2, \qquad \mathrm{Var}(u \mid X) = \sigma^2 I_n.$$

Residual plots can reveal a widening or narrowing spread as fitted values or regressors change. What happens to OLS inference if the conditional error variance is not constant?

The route through heteroskedasticity

1. Derive what unequal error variance changes for OLS estimates and inference.
2. Use residuals to test whether a constant variance is plausible.
3. If a variance pattern can be modeled, use weighted least squares.

For given x_i, $\mathbb{E}(y_i \mid x_i) = x_i'\beta$ is the average outcome; $\mathrm{Var}(y_i \mid x_i) = \mathrm{Var}(u_i \mid x_i)$ is its dispersion around that average.

Learning goals

By the end of this lecture, you should be able to:

- explain the OLS covariance formula with heteroskedastic errors;
- compute and interpret heteroskedasticity-robust standard errors;
- explain the Breusch–Pagan and White tests for heteroskedasticity;
- explain how a variance model produces weighted least squares estimates.

Recall: multiple regression under homoskedasticity

Let $Y = X\beta + u$, with n observations and $k+1$ coefficients. If X has full column rank,

$$\hat{\beta} = (X'X)^{-1}X'Y.$$

Under $\mathbb{E}(u \mid X) = 0$, $\mathbb{E}(\hat{\beta} \mid X) = \beta$.

If also $\mathrm{Var}(u \mid X) = \sigma^2 I_n$, then

$$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X'X)^{-1}, \qquad \hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n - (k+1)}.$$

These are the variance formulas whose assumptions we now examine.

What conditional normality adds

If $u \mid X \sim N(0, \sigma^2 I_n)$, then

$$\hat{\beta} \mid X \sim N\big(\beta, \sigma^2 (X'X)^{-1}\big).$$

Writing $P = X(X'X)^{-1}X'$ gives

$$\hat{u} \mid X \sim N\big(0, \sigma^2 (I_n - P)\big), \qquad \frac{[n-(k+1)]\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-(k+1)}.$$

$\hat{\beta}$ and $\hat{\sigma}^2$ are independent, which supports exact finite-sample t and F inference.

Without normality: large-sample inference

Keep $\mathbb{E}(u_i \mid x_i) = 0$, $\mathrm{Var}(u_i \mid x_i) = \sigma^2$, independent sampling, full rank, and suitable moments. Normal errors are not required for

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}), \qquad Q = \mathbb{E}(x_i x_i').$$

Consequently, conventional t statistics approach $N(0,1)$ and qF approaches χ^2_q for q restrictions. These are asymptotic statements; the normal model gave exact ones.

Heteroskedasticity: allow unequal conditional variances

Replace the common variance by

$$\mathrm{Var}(u_i \mid x_i) = \sigma_i^2 = \sigma^2(x_i).$$

The variance can depend on regressors, and it can also depend on variables absent from the conditional mean equation.

We continue to maintain $\mathbb{E}(u_i \mid x_i) = 0$ and use large-sample results unless we explicitly assume a distribution.

Heteroskedasticity: what survives, and what changes?

Under zero conditional mean and the usual sampling and rank conditions:

- OLS remains unbiased conditionally on X and consistent.
- The homoskedastic standard-error formula is generally wrong, so its t and F conclusions can be wrong.
- OLS loses the Gauss–Markov efficiency guarantee.

Heteroskedasticity concerns the spread of the error, not its conditional mean.

Heteroskedasticity: why OLS remains unbiased

Substitute $Y = X\beta + u$ into the OLS estimator:

$$\hat{\beta} = \beta + (X'X)^{-1}X'u.$$

Therefore, if $\mathbb{E}(u \mid X) = 0$,

$$\mathbb{E}(\hat{\beta} \mid X) = \beta + (X'X)^{-1}X'\mathbb{E}(u \mid X) = \beta.$$

This argument uses the conditional mean restriction. It does not use a common error variance.

Heteroskedasticity: the asymptotic covariance with unequal variances

The OLS error is an average of observation-level terms:

$$\sqrt{n}(\hat{\beta} - \beta) = Q_n^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} x_i u_i, \qquad Q_n = \frac{X'X}{n}.$$

Since $\mathbb{E}(u_i \mid x_i) = 0$, $\mathbb{E}(x_i u_i) = 0$ and

$$\begin{aligned} \operatorname{Var}(x_i u_i) &= \mathbb{E}\big[u_i^2 x_i x_i'\big] = \mathbb{E}\big[x_i x_i' \mathbb{E}(u_i^2 \mid x_i)\big] \\ &= \mathbb{E}\big[\sigma^2(x_i) x_i x_i'\big] \equiv \Omega. \end{aligned}$$

For independent observations, the CLT gives $n^{-1/2} \sum_i x_i u_i \xrightarrow{d} N(0, \Omega)$; with $Q_n \xrightarrow{p} Q = \mathbb{E}(x_i x_i')$,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, Q^{-1} \Omega Q^{-1}).$$

Heteroskedasticity: the correct variance in the observed sample

If errors across observations are conditionally uncorrelated, their conditional covariance matrix is

$$\mathrm{Var}(u \mid X) = D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2), \qquad \sigma_i^2 = \mathrm{Var}(u_i \mid X) = \sigma^2(x_i).$$

Since $\hat{\beta} - \beta = (X'X)^{-1}X'u$, the correct conditional variance is

$$\mathrm{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'DX(X'X)^{-1}.$$

The sample analogue of Ω is given by:

$$\Omega_n = \frac{X'DX}{n} = \frac{1}{n}\sum_i \sigma_i^2 x_i x_i'.$$

If we nevertheless use the homoskedastic variance formula, our t statistics can be wrong.

Heteroskedasticity: using the homoskedastic variance

Suppose we still use: $\widehat{\mathrm{Var}}(\hat{\beta} \mid X) = \hat{\sigma}^2 (X'X)^{-1}, \quad \hat{\sigma}^2 = \dfrac{\hat{u}'\hat{u}}{n-(k+1)}.$

Let $P = X(X'X)^{-1}X'$ and $M = I_n - P$. As $\hat{u} = Mu$: $\hat{u}'\hat{u} = u'Mu = \mathrm{tr}(Muu')$.

Since $\mathbb{E}(u \mid X) = 0$, $\mathbb{E}(uu' \mid X) = D$:

$$\mathbb{E}[\mathrm{tr}(Muu') \mid X] = \mathrm{tr}[M\mathbb{E}(uu' \mid X)] = \mathrm{tr}(MD).$$

Therefore,

$$\mathbb{E}(\hat{\sigma}^2 \mid X) = \frac{\mathrm{tr}(MD)}{n-(k+1)} = \frac{\sum_i (1-P_{ii})\sigma_i^2}{n-(k+1)}.$$

Moreover, $\sum_i P_{ii} = \mathrm{tr}(P) = \mathrm{tr}[(X'X)^{-1}X'X] = k+1$.

If $\sigma_i^2 = \sigma^2$ for all i, then $\hat{\sigma}^2$ is conditionally unbiased for σ^2 and consistent.

Heteroskedasticity: using the homoskedastic variance

With unequal variances, fixed k, and suitable regularity conditions, $\hat{\sigma}^2$ still converges to the *average* conditional variance:

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \sigma^2(x_i) = \bar{\sigma}^2 = \mathbb{E}[\sigma^2(x_i)], \qquad \hat{\sigma}^2 \xrightarrow{p} \bar{\sigma}^2.$$

However, we have used $\hat{\sigma}^2 Q_n^{-1}$ to estimate the covariance of $\sqrt{n}(\hat{\beta} - \beta)$, while the correct conditional covariance is $Q_n^{-1}\Omega_n Q_n^{-1}$. The required covariance depends on where the larger variances occur among the regressor values. The scalar average does not contain that information.

The difference is

$$Q_n^{-1}\left[\frac{1}{n}\sum_i (\sigma_i^2 - \hat{\sigma}^2)x_i x_i'\right] Q_n^{-1}.$$

If variance depends only weakly on regressor values, the discrepancy may be small; stronger dependence can make the resulting tests misleading.

Inference is the immediate problem

When variance depends on x_i, the wrong covariance estimate enters

$$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{\mathrm{se}(\hat{\beta}_j)}.$$

An OLS coefficient can still target the correct population slope while a conventional standard error gives a misleading test or confidence interval.

First response: keep OLS coefficients and correct the covariance estimate.

Estimate the unequal variance contributions

The population middle matrix is $\Omega = \mathbb{E}(u_i^2 x_i x_i')$, but the errors u_i are unobserved. OLS residuals satisfy

$$\hat{u}_i = u_i - x_i'(\hat{\beta} - \beta).$$

Since $\hat{\beta} \xrightarrow{p} \beta$, for a fixed observation i, $\hat{u}_i^2 \xrightarrow{p} u_i^2$. This is the realized squared error, not the conditional variance $\sigma_i^2 = \mathbb{E}(u_i^2 \mid x_i)$.

Under suitable moment conditions, their weighted sample average satisfies

$$\hat{\Omega}_n = \frac{1}{n} \sum_i \hat{u}_i^2 x_i x_i' \xrightarrow{p} \Omega = \mathbb{E}(u_i^2 x_i x_i').$$

Across observations, the average of $\hat{u}_i^2 x_i x_i'$ converges to the population average Ω.

Heteroskedasticity-robust covariance

A heteroskedasticity-robust estimate of the OLS covariance is

$$\widehat{\mathrm{Var}}(\hat{\beta}) = (X'X)^{-1} \left(\sum_i \hat{u}_i^2 x_i x_i' \right) (X'X)^{-1}.$$

Use its diagonal elements for robust standard errors. With independent observations and suitable large-sample conditions,

$$\frac{\hat{\beta}_j - \beta_{j,0}}{\sqrt{[\widehat{\mathrm{Var}}(\hat{\beta})]_{jj}}} \xrightarrow{d} N(0,1).$$

Remark. Robust large-sample inference works with either homoskedastic or heteroskedastic errors. Under homoskedasticity and conditional normality, the usual variance formula with Student critical values gives exact finite-sample inference.

Heteroskedasticity-robust LM statistic

If software does not provide robust covariance estimates, use this test.

Consider a model with potentially heteroskedastic errors:

$$Y_i = x_i'\beta + z_i'\delta + u_i.$$

We test $H_0 : \delta = 0$ against $H_a : \delta \neq 0$, where $z_i \in \mathbb{R}^q$.

1. Regress Y on X by OLS and save the residuals $\hat{u}_i$.
2. Regress each of the q excluded regressors on X; save its residuals $\hat{r}_{ij}$.
3. Regress the constant 1 on the q products $\hat{u}_i\hat{r}_{ij}$, without an intercept.
4. If SSR_1 is the final residual sum of squares, then $n - \text{SSR}_1 \xrightarrow{d} \chi^2_q$ under H_0.

Idea: Under H_0, each product $\hat{u}_i\hat{r}_{ij}$ averages close to zero.
The final regression uses their dispersion, allowing unequal error variances.

Testing for Heteroskedasticity

Breusch–Pagan: model the variance pattern

Consider the usual linear regression model under MLR.1–MLR.4:

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + u_i.$$

We test MLR.5 (homoskedasticity): $H_0 : \mathrm{Var}(u_i \mid x_i) = \sigma^2$.

Because $\mathbb{E}(u_i \mid x_i) = 0$, this is equivalent to: $H_0 : \mathbb{E}(u_i^2 \mid x_i) = \mathbb{E}(u_i^2) = \sigma^2$.

Under H_a, $\mathbb{E}(u_i^2 \mid x_i)$ may depend on the regressors in many ways. Consider a linear pattern, with $z_i = (x_{i1}, \ldots, x_{ik})'$:

$$u_i^2 = \alpha_0 + z_i'\alpha + v_i, \qquad \mathbb{E}(v_i \mid z_i) = 0.$$

In this auxiliary model, $H_0 : \alpha = 0$ versus $H_a : \alpha \neq 0$.

If u_i^2 were observed, an F or LM statistic could test H_0.

Breusch–Pagan: the auxiliary regression

We can estimate the equation

$$\hat{u}_i^2 = a_0 + z_i' a + e_i$$

and compute the F or LM statistics for the joint significance of the regressors in z_i.

Then, the F statistic is

$$F = \frac{R^2_{\hat{u}^2}/k}{(1 - R^2_{\hat{u}^2})/(n-k-1)},$$

which has approximately an $F_{k,n-k-1}$ distribution under H_0.

The LM statistic is given by

$$LM = nR^2_{\hat{u}^2} \xrightarrow{d} \chi^2_k \quad \text{under } H_0.$$

If the Breusch–Pagan test gives a sufficiently small p-value, use heteroskedasticity-robust standard errors or weighted least squares estimation.
If variance may depend on only q regressors, include those in z_i and replace k by q throughout the statistics and reference distributions.

White: test a broader variance pattern

To detect variance patterns beyond a linear relationship with the regressors, regress $\hat{u}_i^2$ on the original regressors, their squares, and their pairwise products.

$$\hat{u}_i^2 = \alpha_0 + \sum_j \alpha_j x_{ij} + \sum_j \gamma_j x_{ij}^2 + \sum_{j<\ell} \delta_{j\ell} x_{ij} x_{i\ell} + v_i.$$

$H_0 : \alpha_j = \gamma_j = \delta_{j\ell} = 0$ for all j, ℓ (homoskedasticity).

H_1 : at least one nonconstant auxiliary coefficient is nonzero.

With q nonconstant terms, $LM = nR^2_{\text{aux}} \xrightarrow{d} \chi^2_q$ under H_0. Squares and cross-products can detect patterns missed by a linear variance regression, but the number of terms grows quickly with k.

White: a compact fitted-value version

The full White regression can have many terms: k coefficients for squared regressors and $k(k-1)/2$ for their cross-products.

Remember that fitted values are linear functions of the independent variables:

$$\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^{k} \hat{\beta}_j x_{ij}.$$

Squaring this equation produces squares and cross-products of the regressors.

Therefore, we test for heteroskedasticity by estimating

$$\hat{u}_i^2 = \alpha_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + v_i.$$

Test $H_0 : \delta_1 = \delta_2 = 0$ against H_1 : at least one is nonzero, using $nR^2_{\text{aux}} \xrightarrow{d} \chi^2_2$ under H_0.

Weighted least squares

Weighted least squares: remove the variance pattern

Suppose heteroskedasticity is known up to a multiplicative constant, with known h_i:

$$Y_i = x_i'\beta + u_i, \quad \mathbb{E}(u_i \mid x_i) = 0, \quad \mathrm{Var}(u_i \mid x_i) = \sigma^2 h_i, \qquad h_i > 0.$$

Divide every term by $\sqrt{h_i}$:

$$\frac{Y_i}{\sqrt{h_i}} = \frac{x_i'\beta}{\sqrt{h_i}} + \frac{u_i}{\sqrt{h_i}}.$$

Where the transformed error has conditional moments:

$$\mathbb{E}\left(\frac{u_i}{\sqrt{h_i}}\middle| x_i\right) = 0, \qquad \mathrm{Var}\left(\frac{u_i}{\sqrt{h_i}}\middle| x_i\right) = \sigma^2.$$

The transformed equation has a constant conditional error variance.

Weighted least squares is OLS after transformation

OLS on the transformed equation solves

$$\min_b \sum_{i=1}^{n} \frac{(Y_i - x_i'b)^2}{h_i}.$$

Hence

$$\hat{\beta}_{WLS} = \left(\sum_i \frac{x_i x_i'}{h_i} \right)^{-1} \left(\sum_i \frac{x_i Y_i}{h_i} \right).$$

Observations with larger conditional variance receive smaller weight. If the h_i are known and the mean assumptions hold, WLS has the Gauss–Markov efficiency property for the transformed model. This is generalized least squares (GLS) with a diagonal error covariance matrix.

When the variance factor is unknown

Often, the function $h(x_i)$ is not obvious. In many cases, we can model h and use the data to estimate the unknown parameters in this model.

To this end, consider $h_i = h(z_i, \theta) > 0$, with unknown θ and variance predictors z_i.

We model heteroskedasticity assuming that

$$\mathrm{Var}(u_i \mid x_i) = \sigma^2 h(z_i, \theta) = \sigma^2 \exp(z_i'\theta).$$

There is no intercept in $z_i'\theta$. It would multiply every h_i by the same constant; σ^2 absorbs it.

z_i can include a subset of x_i, or other relevant variance predictors.

To estimate θ, we will transform this equation into a linear form that can be estimated by OLS.

Estimate a log variance relationship

To estimate h_i, write each squared error as its modeled variance times a positive random factor:

$$u_i^2 = \sigma^2 \exp(z_i'\theta)\eta_i, \qquad \mathbb{E}(\eta_i \mid z_i) = 1.$$

Assume η_i is independent of z_i. Taking logs then gives

$$\log(u_i^2) = c + z_i'\theta + e_i, \quad c = \log \sigma^2 + \mathbb{E}(\log \eta_i), \quad \mathbb{E}(e_i \mid z_i) = 0.$$

Under the usual sampling and rank conditions, it satisfies the Gauss–Markov assumptions.

If u_i^2 were observed, OLS estimates of θ would be unbiased.

Feasible GLS: estimated weights

Because u_i is unobserved, use $\hat{u}_i$ instead and recover the fitted values $\hat{g}_i$ from the regression of $\log(\hat{u}_i^2)$ on z_i.

Then $\hat{h}_i = \exp(\hat{g}_i)$. We use WLS with weights $1/\hat{h}_i$ in place of $1/h_i$.

Equivalently, fit the transformed model

$$\frac{Y_i}{\sqrt{\hat{h}_i}} = \frac{x_i'\beta}{\sqrt{\hat{h}_i}} + \frac{u_i}{\sqrt{\hat{h}_i}}.$$

This is feasible generalized least squares (FGLS), specifically feasible WLS for a diagonal error covariance matrix.

The intercept in $\hat{g}_i$ rescales every $\hat{h}_i$ equally and therefore does not change the WLS coefficient estimates.

Feasible GLS: the five steps

1. Regress Y_i on x_i by OLS and save the residuals $\hat{u}_i$.
2. Compute $\log(\hat{u}_i^2)$ for each observation.
3. Regress $\log(\hat{u}_i^2)$ on z_i and save fitted values $\hat{g}_i$.
4. Set $\hat{h}_i = \exp(\hat{g}_i) > 0$.
5. Refit the mean equation by WLS using weights $1/\hat{h}_i$; equivalently, run OLS on $Y_i/\sqrt{\hat{h}_i}$ and $x_i/\sqrt{\hat{h}_i}$.

Estimating h_i makes the procedure feasible; its efficiency gain relies on the variance specification.

What to take from the analysis

- Heteroskedasticity can leave OLS slopes intact while invalidating conventional standard errors.
- Robust covariance estimates repair large-sample OLS inference without specifying a variance function.
- Residual-based tests look for specific patterns of changing variance.
- Weighted least squares uses a modeled variance pattern to change the estimates and can gain precision when that pattern approximates the true conditional variance well.